

WIRELESS NETWORKING

Day – 8 Switch and router theory.

Task – 1 Cisco packet Tracer and network simulation.

- Here is the quick, step-by-step CLI method to create both VLANs:
- Click the Switch in Packet Tracer and open the CLI tab.
- Press Enter, then type these commands:

```
- enable
  configure terminal
  vlan 10
  name Sales
  vlan 20
  name Support
  end
```

- Type `show vlan brief` to verify that 10 Sales and 20 Support appear in your active VLAN list.

Task – 2 Different ports on VLAN 10.

- To assign the ports efficiently and verify the setup in Cisco Packet Tracer, open the Switch CLI tab and enter these commands:
- Step 1: Assign Ports to VLANs

```
- enable
  configure terminal

  ! Assign FastEthernet 0/1 to 0/3 to VLAN 10 (Sales)
  interface range fastEthernet 0/1 - 3
  switchport mode access
  switchport access vlan 10
  exit
```

```

! Assign FastEthernet 0/4 to 0/5 to VLAN 20 (Support)
interface range fastEthernet 0/4 - 5
switchport mode access
switchport access vlan 20
end

```

- Step 2: Verify Membership
- Run `show vlan brief` in Privileged EXEC mode (Switch#).
- Expected Output:

```
Switch# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8...
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5

Task – 3 Connecting two device using ethernet link.

- Here is the simplest summary of the complete 3-step setup to copy directly into Cisco Packet Tracer:
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- 1. Configure Router 1
- Click Router1 > CLI and run:

```

enable
configure terminal
interface g0/0/0
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
ip route 192.168.20.0 255.255.255.0 10.0.0.2
end

```

- 2. Configure Router 2
- Click Router2 > CLI and run:

```
- enable
  configure terminal
  interface g0/0/0
    ip address 10.0.0.2 255.255.255.252
    no shutdown
  exit
  interface g0/0/1
    ip address 192.168.20.1 255.255.255.0
    no shutdown
  end
```

- Type ping 192.168.20.1 on Router 1. You will see Success rate is 80 or 100 percent, confirming the connection works!

Task – 4 Scenario based on VLAN.

- 1. Simulation Setup & Ping Failure Test :- PC-Sales (VLAN 10): IP 192.168.10.10 /24 | Default Gateway: 192.168.10.1 | Port: Fa0/1
- PC-Support (VLAN 20): IP 192.168.20.20 /24 | Default Gateway: 192.168.20.1 | Port: Fa0/4
- On PC-Sales, open the Command Prompt and type:
- Result: Request timed out.
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- 2. Why the Ping Fails :- Layer 2 Broadcast Isolation: VLANs break up a switch into completely separate logical networks (broadcast domains). Switch ports assigned to VLAN 10 cannot forward Layer 2 frames to ports in VLAN 20.
- Lack of Layer 3 Routing: Pinging across different IP subnets (192.168.10.x to 192.168.20.x) requires a Layer 3 device (a Router or Layer 3 Switch) to route packets between the subnets. Without one, traffic cannot cross the VLAN boundary.
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- 3. Solution: Router-on-a-Stick Configuration :- Connect the switch port Fa0/24 to the Router port g0/0/0 with a straight-through cable.
- Step A: Configure Switch Trunk Port
- Step B: Configure Router Sub-interfaces

- 4. Verification :- Re-run ping 192.168.20.20 on PC-Sales. You will see Reply from 192.168.20.20: bytes=32 time<1ms TTL=127, confirming successful Inter-VLAN communication.
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